Accelerating DORA Compliance: Automating Network Access Control at Scale with Infrastructure as Code

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Marco Basso
Consulting Engineer @ Cisco

Luca Lorenzo Dell'Orto
Network Engineer @ Intesa Sanpaolo

Agenda

Intesa Sanpaolo & Cisco

Intesa at a Glance

Banking & Investments

Intesa Sanpaolo provides banking services for individuals and businesses, including loans, accounts, payments, investment products, portfolio management, and insurance solutions.

Digital & Innovation

Intesa Sanpaolo provides advanced online and mobile banking services, leveraging digital platforms and fintech solutions to enhance efficiency and customer experience.

National Infrastructure

With over 2,500 branches and more than 8,000 network devices, Intesa Sanpaolo ensures comprehensive coverage and reliable connectivity across Italy.

Meet the Teams

Luca Lorenzo Dell'Orto

Marco Basso

Cisco

Andrea Bertorello
Andrea Vitali
Antonio Piepoli
Armando Garcia
Cristian Garotta
Fabio Salvemini
Lara Gatti
Lorenzo Botto
Marco Basso
Mauro Gambaretto
Paola Mattioli
Stefano Tadiello

Intesa Sanpaolo

Daniele Mastroti
Davide Rendiniello
Giorgio Roberto
Luca Lorenzo Dell'Orto
Michele Biroli
Stefano Rossi

Partnership History

Strong and Intense

Giorgio Roberto

From Delivery to Partnership – A Journey of Trust

- Partnership Intesa Sanpaolo & Cisco Professional Services established in 2000
- Long-standing collaboration and joint delivery model accelerating IT infrastructure digitalization through Cisco cutting-edge technologies
- Proven impact across Data Center, Enterprise, Collaboration, Security, Assurance
- Advancing the journey toward Infrastructure as Code (IaC), with network infrastructure modeled for consistent, scalable deployment and operations
- Two ongoing projects leveraging IaC: ACI and Catalyst Center to deliver model-driven, scalable network operations
- So, what's next? Redefining network infrastructure operations and control with AIOps

DORA & Compliance

DORA – Redefining ICT Risk Management Across Finance

What is DORA?

The Digital Operational Resilience Act (DORA) is a European Union regulation requiring financial entities to enhance their digital operational resilience, including the security of their ICT systems and third-party providers.

Steps to Compliance

To comply with regulatory requirements, it became necessary to implement a Network Access Control solution across the entire perimeter, restricting network access to authorized devices that meet security compliance standards.

Phased Rollout

The project will roll out the solution in phases over about a year, implementing controls on more than 8,000 switches across 2,500 branches.

Business Drivers

"When automation is done right, it transforms operations from a cost center into a scalable growth engine

Cisco Community

Key Business Delivery Priorities

OPEX Budget Cap

Time Constraints

DORA Compliance

Solution Overview

Key Technical Challenges

Manual Workflows

Lack of Standard Configuration

No Input Validation

No Periodic Testing & Reporting

Deadline Constraints

Lack of Version Control

Intesa Sanpaolo Network Infrastructure

Geographic Network

- 2,500+ sites across 20 Italian region
- 8,000+ Network Access Devices (NAD) across 10 product types
- Catalyst Center XL Cluster

ISE Cluster

- Cluster distributed across 2 locations
- 8 server nodes responsible for handling access requests
- Authentication via certificates and profiling of connected devices

Secured Assets

- Employee devices (e.g., PCs, laptops)
- Security devices (e.g., IP cameras, badge readers)
- Business-related IT (e.g., ATMs, cash registers)
- Collaboration devices (e.g., IP phones, video conferencing)

Digital Architecture with Network as Code

Network as Code

Implementation and Operation

Design

Code Repository | Pre-checks | CI/CD

Validation/Test

Requirements

Cisco Customer Experience

Data Model - Per Architecture Best Practices

Documentation

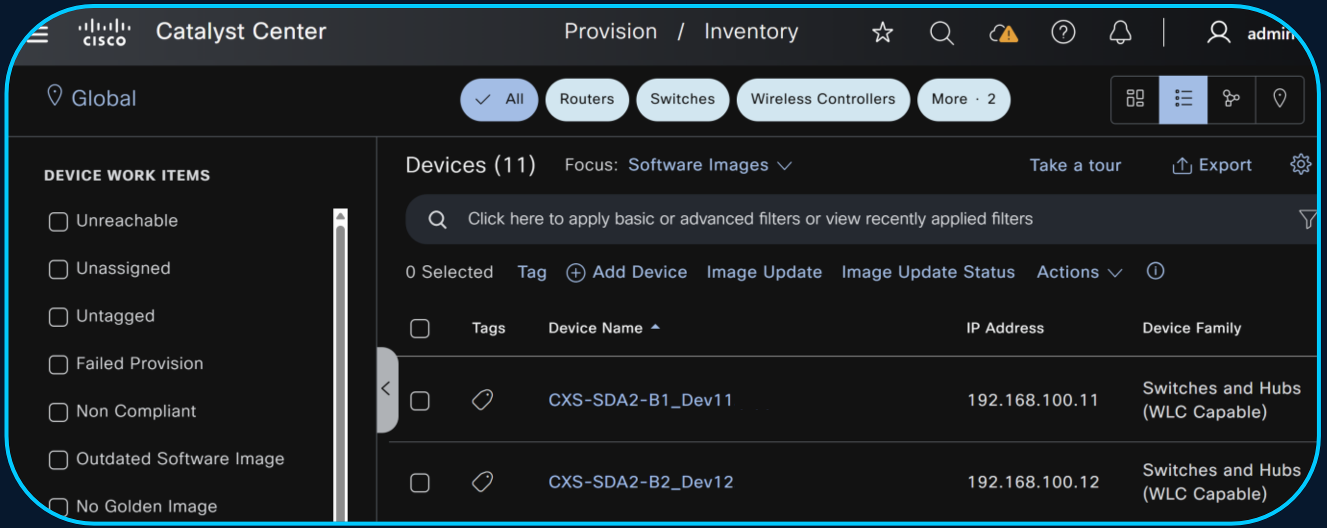
Operations

Infrastructure Adapter – Terraform Provider, Ansible, Open Tofu, ...

Digital infrastructure Interaction – API

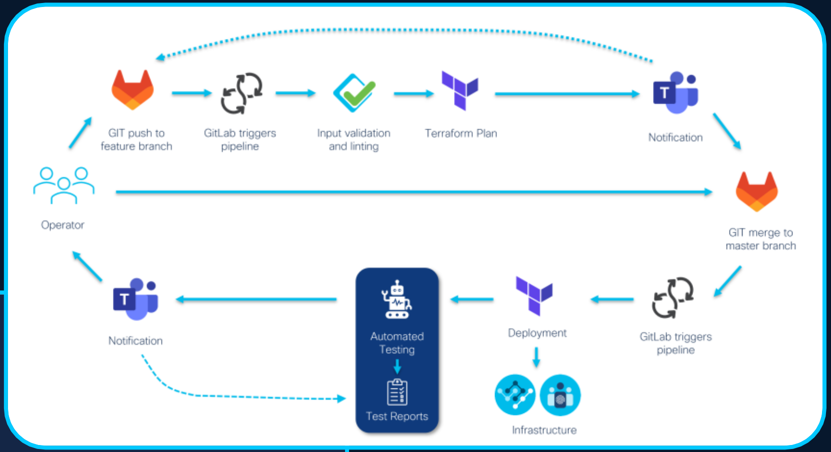
ACI | NDFC | SDWAN | ISE | Firewall | Meraki | Catalyst Center | Unified Branch | IOS-XR | IOS-XE | NX-OS | Thousand Eyes | 3rd parties

Automated Infrastructure Management Processes



Old Style

New Style




```yaml
catalyst_center:
  inventory:
    devices:
      - name: PNP_DEVICE
        fqdn_name: PNP_DEVICE.cisco.eu
        device_ip: 192.158.142.20
        serial_number: FAB12345602
        pid: C9300-48P
        state: PROVISION
        device_role: BORDER ROUTER
        site: Global/Poland/Krakow/Bld A
```

YAML

Not just a shift in technology, but an innovation across a whole set of processes

Network as Code

How We Solved Intesa Business Problem?

NAC (Network Access Control) vs NaC (Network as Code)

Rundeck

Orchestrator used to define workflows composed of atomic operations to create a pipeline that performs validation, implementation, and testing.

Catalyst Center

Management and automation platform, leveraged to automate device inventory, manage configuration versions, and deploy configuration changes through APIs integrated with Terraform.

Terraform

IaC tool integrated with Catalyst Center to implement infrastructure changes by applying updates to YAML-based data models, ensuring consistency, automation, and repeatability.

Network as Code Tool

Open-source tools were leveraged throughout the workflows to validate the data model, ensure data integrity, and test the applied configuration for accuracy and reliability.

Solution Technology Stack

CX Customized

CX LCS

CX Open Source

CX LCS Community

Automated Testing

Pre-Change Validation

Rundeck Orchestrator

Version Control System

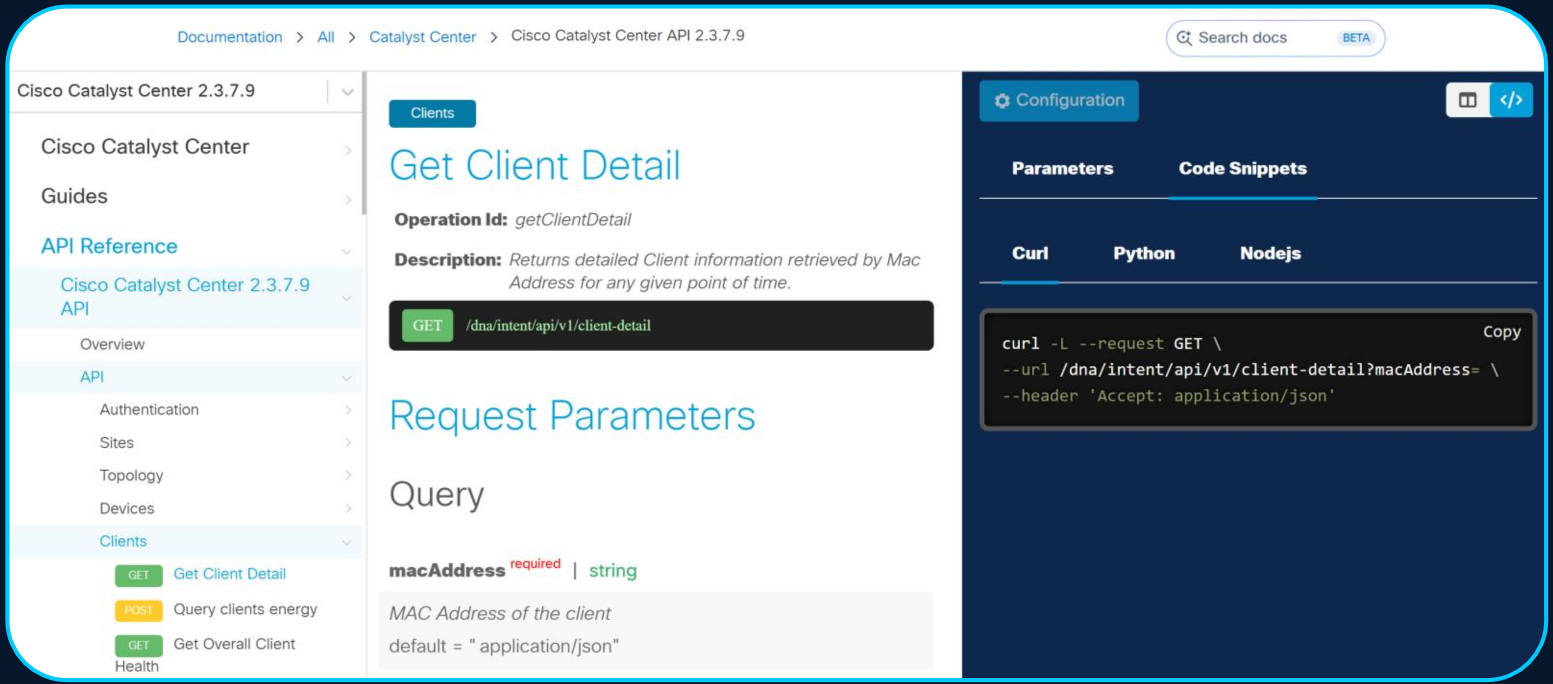
Data Model (NaC)

Catalyst Center Terraform Provider

Catalyst Center API

IaC Stack – Catalyst Center API

Public Documentation



APIs provide a programmatic configuration interface and expose specific capabilities of the Catalyst Center platform.

IaC Stack – Terraform

[Public GitHub Terraform Module Repository]

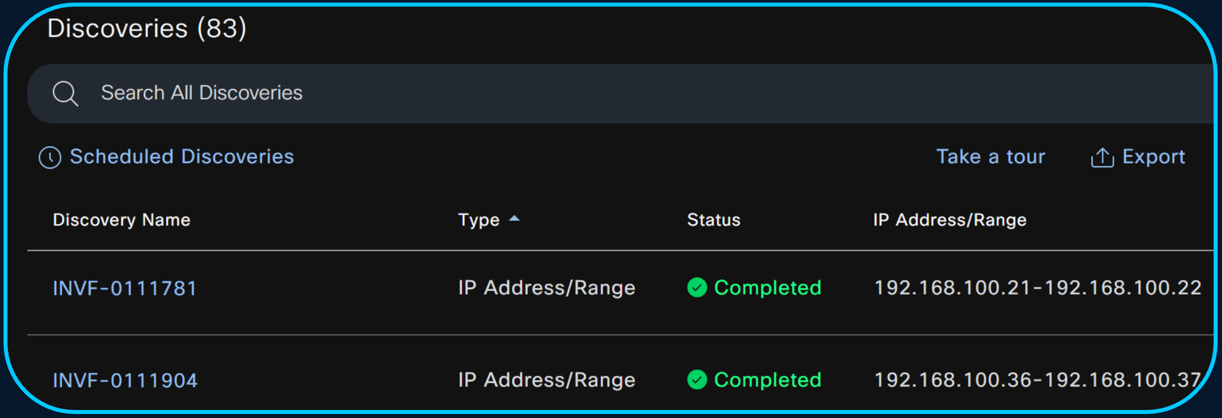
[Public GitHub Terraform Provider Repository]

- Terraform is an Infrastructure Resources Manager
- Applies desired changes through the Catalyst Center APIs using its dedicated provider
- Multi-State Architecture for Improved Delivery Efficiency and Scalability

```
.
├── catalyst-center-region-Abruzzo
│   └── main.tf
├── catalyst-center-region-Basilicata
│   └── main.tf
├── catalyst-center-region-Calabria
│   └── main.tf
└── catalyst-center-region-Veneto
    └── main.tf
```

TF Multi-State Hierarchy Model

IaC Stack – NaC Data Model



NaC Data Model Documentation

<branch-ID>.nac.yaml

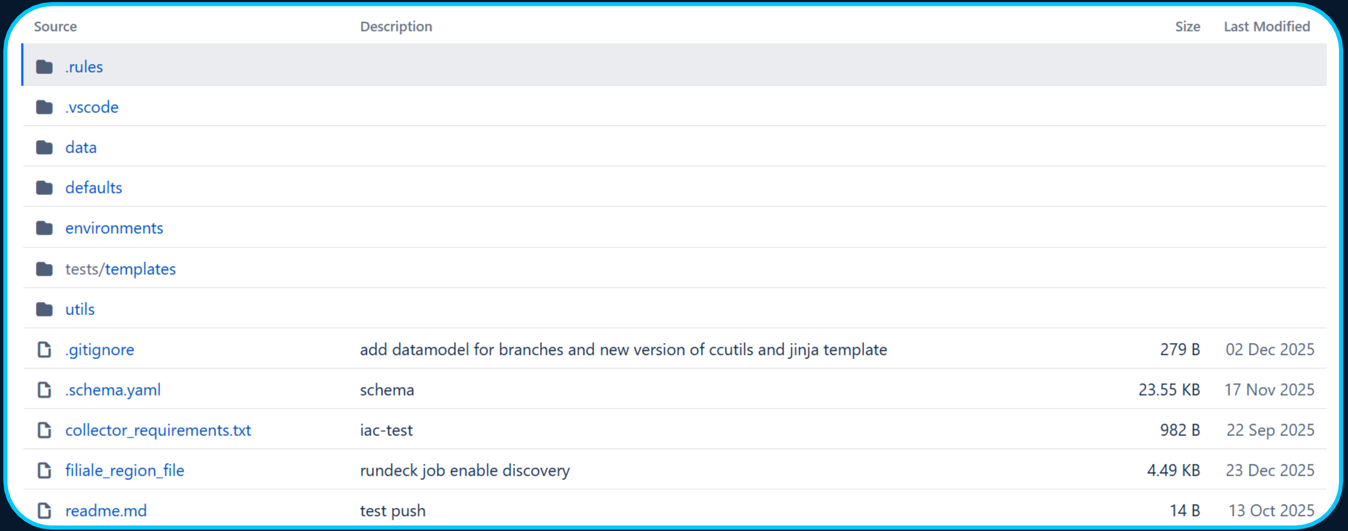
```yaml
catalyst_center:
    inventory:
        discovery:
        - name: <branch-ID>
          type: Multi Range
          global_credential_list:
          - dnacadmin
          ip_address_list: 10.1.2.1-10.1.2.5
          protocol_order: SSH
```

- NaC enables the Catalyst Center to be described through an open-source model
- The model is defined using the YAML human-readable language

IaC Stack – Version Control System

- The data model is maintained in a version-controlled repository
- All changes to the data model, including configuration updates, are automatically tracked and saved
- Provides one-click support for configuration rollback

Bitbucket GUI

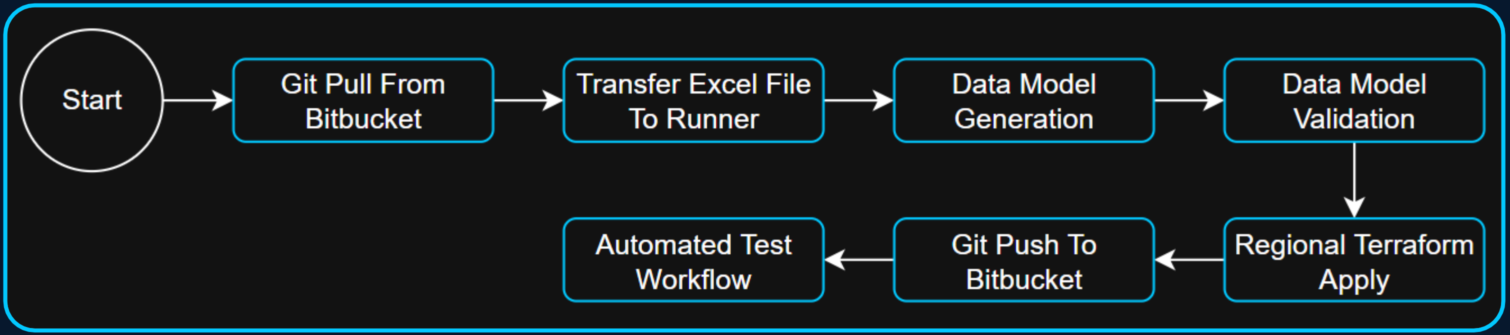


Source	Description	Size	Last Modified
.rules			
.vscode			
data			
defaults			
environments			
tests/templates			
utils			
.gitignore	add datamodel for branches and new version of ccutils and jinja template	279 B	02 Dec 2025
.schema.yaml	schema	23.55 KB	17 Nov 2025
collector_requirements.txt	iac-test	982 B	22 Sep 2025
filiale_region_file	rundeck job enable discovery	4.49 KB	23 Dec 2025
readme.md	test push	14 B	13 Oct 2025

IaC Stack – Rundeck Orchestrator

- Orchestrator for executing custom workflows
- Workflows consist of an ordered sequence of atomic jobs, each designed to perform a specific function
- Atomic jobs are executed by a dedicated slave (linux machine), ensuring reliable and orderly workflow execution

Automated Graphical Output in Rundeck – Generic Workflow



IaC Stack – Pre-Change Validation (NaC)

Data Model Validation

- Validation of the data model is essential to ensure the correct configuration of the Catalyst Center
- CLI tool for validating NaC Data Model: iac-validate
- Enables checks on format, syntax, semantics, and custom compliance
- Enables the creation of specific rules → Intesa use case: verifying that branch ID codes follow a defined structure:
  - INVF-1234567_ABC123_Fil
  - INTL-12345_ABC123_Fil

How to install it?

```
runner@cc-auto:~/Intesa/intesa-as-code$ pip install iac-validate
```

IaC Stack – Automated Testing (NaC)

CLI Tool for Testing

- Post-implementation automated testing ensures that changes have been completed as expected
- CLI tool for validating NaC Data Model: iac-test
- Definition of specific tests based on project requirements:
  - Pre-migration: Verify device connectivity from the Catalyst Center and ensure software version compliance
  - Post-migration: Confirm successful NAC activation

Statistics by Suite	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Report Totale	3324	3213	62	49	01:26:50	
Report Totale . Oper	3324	3213	62	49	01:00:15	
Report Totale . Oper . Device Assignment	660	647	0	13	00:37:49	
Report Totale . Oper . Device Reachability	660	647	13	0	00:37:34	
Report Totale . Oper . Device Software Version	660	635	12	13	00:37:18	
Report Totale . Oper . Discovery	684	653	31	0	00:10:50	
Report Totale . Oper . Dot1X	660	631	6	23	01:00:15	

Tests are summarized in detail in a report accessible via web browsers (Chrome, Firefox, etc.).

What has been developed by putting all these components together?

Custom Workflows: Tailored Based on Use Cases

The requirements defined by Intesa served as the foundation for building a fully customized solution that meets the delivery requirements.

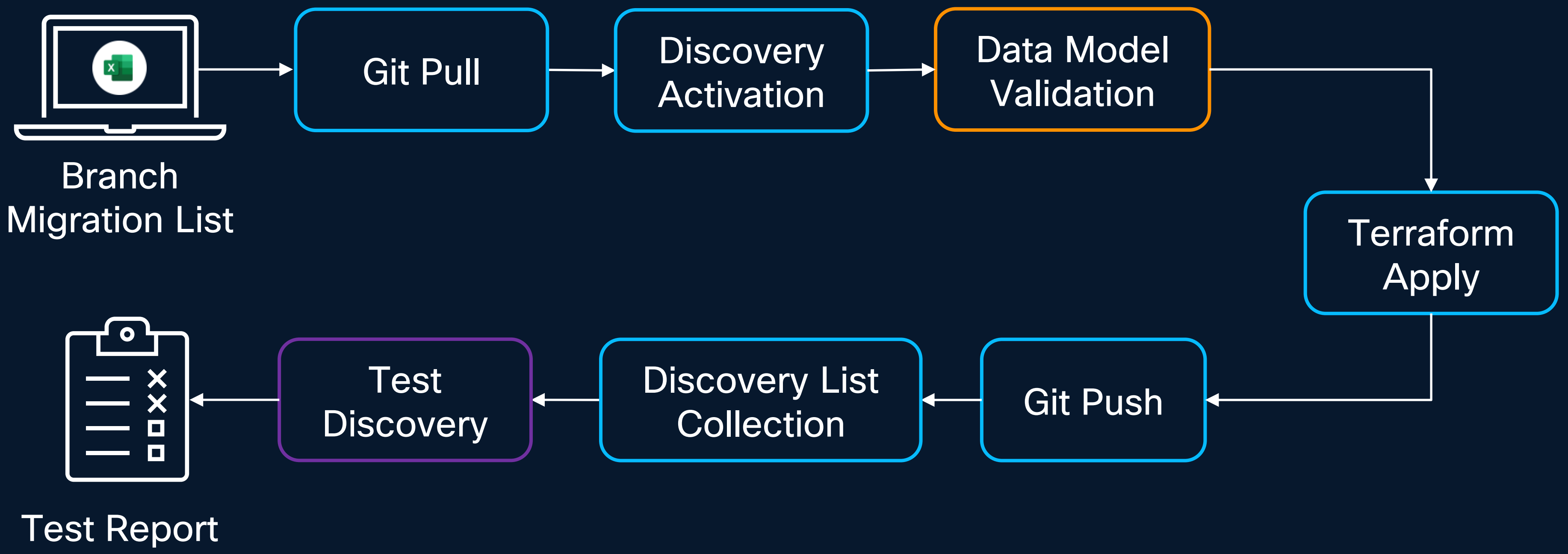
The solution consists of 3 workflows. By executing them in sequence, the NAC is activated for a specific branch, provided as input.

1. Device Discovery: Network device onboarding for centralized Catalyst Center management
2. Site Device Assignment: Assignment of individual network devices to their respective sites through centralized provisioning, enabling full management of the devices
3. Branch Migration: Using the device’s existing configuration, a NAC activation template is generated and deployed via the Catalyst Center to all devices belonging to the branches specified in the first-phase input list

1st Phase – Device Discovery

Input: A list of branch offices for which NAC activation should be triggered

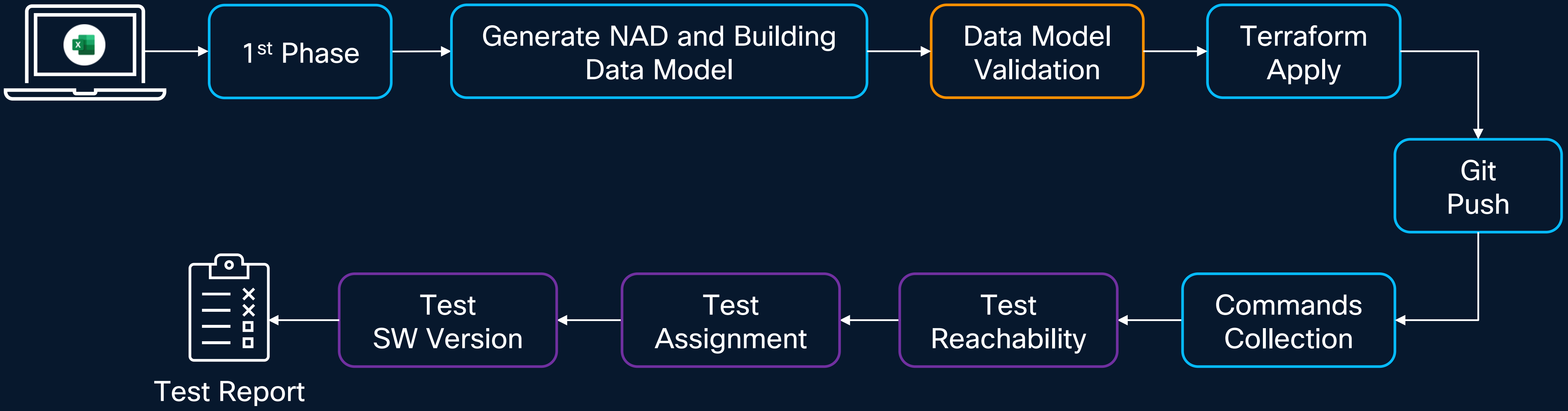
Output: Discovery processes are created in the Catalyst Center for the specified branch offices. A report summarizing the results of the various discoveries is generated automatically



2nd Phase – Site Device Assignment

Input: A list of branch offices for which NAC activation should be initiated as the first phase

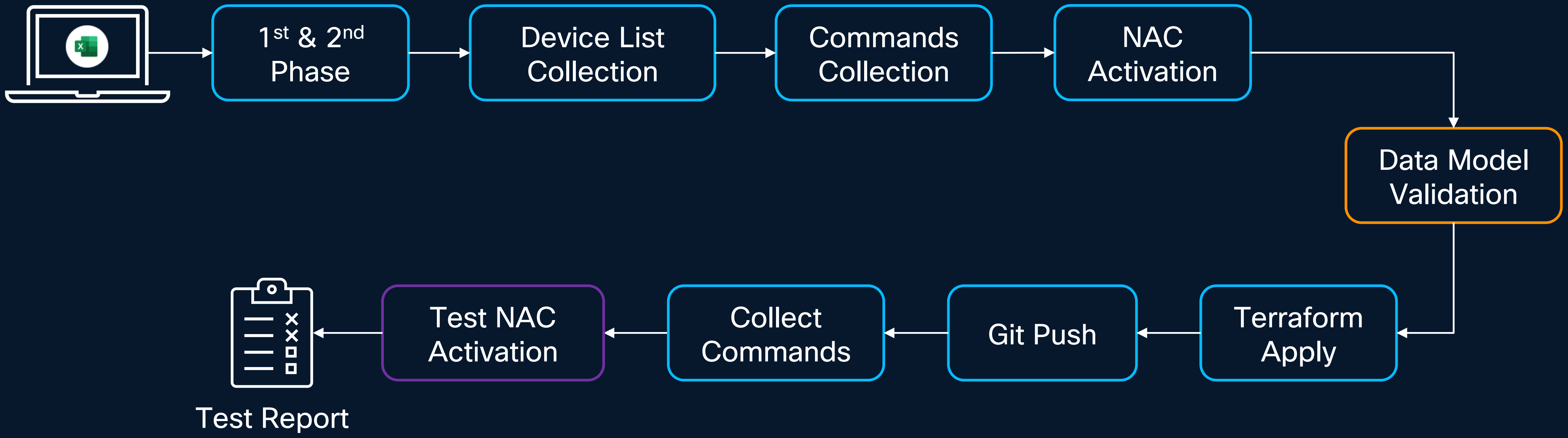
Output: Network devices associated with their respective sites. Report containing final test results



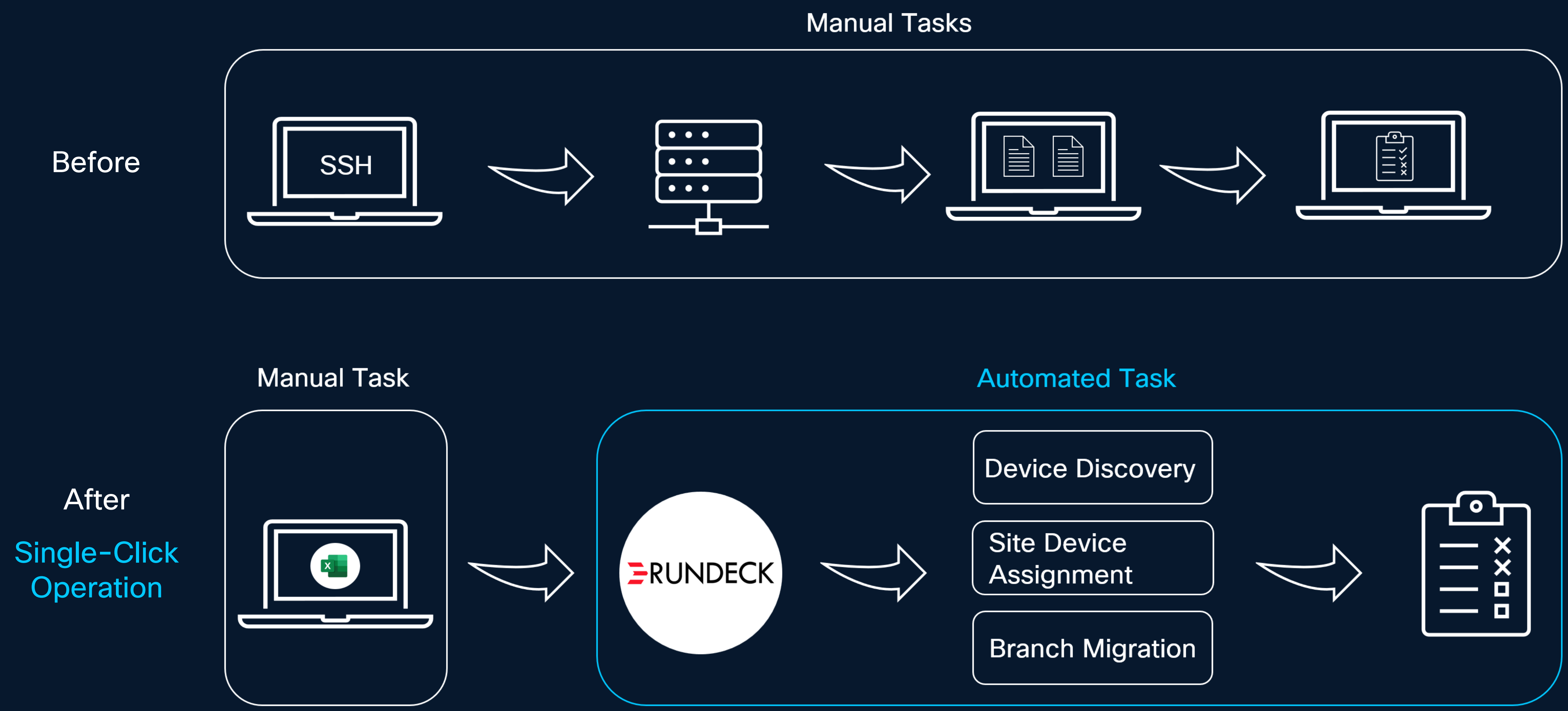
3rd Phase – Branch Migration

Input: A list of branch offices for which NAC activation should be carried out in the first and second phases

Output: All network devices belonging to the input list of sites will have NAC functionality enabled. A report summarizes the final test results



Migration Process Comparison



Value-Driven Benefits

How This Solution Transforms Operations

Reduced by 70%

73 Hours Saved

	With Automation	Without Automation
Average Migration Time per Device *	9 min. **	30 min.
Input Error Handling	Pre-Change Validation	Error-Prone Manual Handling
Ensuring Testing and Compliance	100% Automated with Periodic Report Reviews	Manual Testing Without Automated Reporting
Bulk Configuration Changes	Edit Once, Push Everywhere	Manual Configuration per Device
CC-managed device	Automated Bulk Provisioning	Manual Onboarding and Provisioning
Staffing of Operational Resource	Automation Started with a Single Action	Full-Time Resource Engagement

OPEX Drastically Reduced

* Metric based on a sample of 630 switches, covering all 3 workflow

** Metric performance continues to improve following recent CX-led workflow enhancements

Conclusion

Intesa Sanpaolo & Innovation: Shaping the Future Together

Continuous Innovation

Intesa Sanpaolo has long been, and continues to be, one of the most innovation-driven companies, ready to implement cutting-edge technologies

Strategic Partnership

The collaboration between Intesa Sanpaolo and Cisco is aimed at guiding and facilitating the adoption of these technologies, creating value for the organization

Automation and AI

From process automation to the implementation of artificial intelligence via AIOps, the partnership aims to enhance efficiency in managing and monitoring IT infrastructure

What’s Next?

Data Model-driven Automation

Services as Code

Standardization

NSO

+

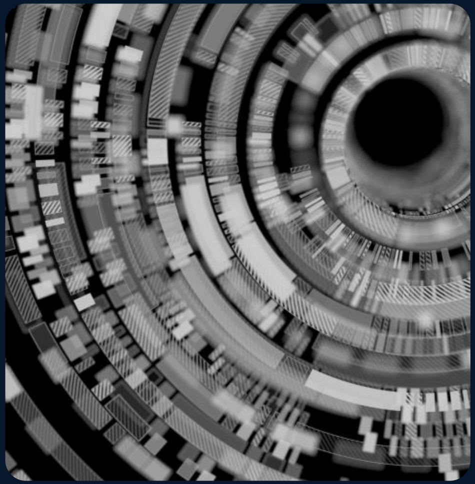
Observability & Assurance

Cisco IQ

Splunk, AppD.

Thousand Eyes

+



Intelligent Decisions rules, ML & AI

Cisco IQ (+AI Agents)

Crosswork

AI Canvas

=

Autonomous Networks AIOps

This is a journey to improve all three areas, starting now!

Thank you